

Exercise 1

Build a regular expression pattern that matches the following rules: 1. Starts with 1–3 occurrences of the character 'S'. 2. Followed by at least 3 occurrences of the substring "com". 3. Followed by 0 or 1 occurrence of the character '%'. 4. Ends with 2 occurrences of the group (@#\$).

Exercise 2

Build a regular expression pattern that matches the following rules: 1. Starts with either "0" or "+84". 2. Followed by exactly 9 digits.

(This models a typical Vietnamese phone number format.)

Exercise 3

Build a regular expression pattern that matches strings of the form:

(local)number

Where: - local is 2–4 digits, starting with 0. - number is 5–7 digits.

Exercise 4

Build a regular expression pattern that matches any string containing 'a' or 'b', but that must **not** contain the substring "ab".

Exercise 5 — Student ID Validation

Write a program that allows the user to enter a student ID, then checks whether the ID matches the required format.

The student ID format is "Bxxxxxxx", where each x is a digit from 1–9. (Use a regular expression to enforce this format.)